Requirements Gathering

Real-Time Chemotherapy Patient Monitoring with AI-Based Anomaly Detection

Scope statement: The prototype monitors chemotherapy patients while they are in the hospital, using synthetic device readings, simulated nursing entries and laboratory events. It combines Kafka/Spark data engineering with Isolation Forest anomaly detection on heart rate and temperature. Reviewers receive traceable, independent rule-based alerts. Full session scheduling, home monitoring, real clinical integrations, autonomous medication changes and clinical diagnosis are not in scope.

1. Identify Stakeholders

Following the supplied process guide, the team identifies who is affected by the system and records both their interest and influence. These are proposed roles in an educational simulation, not people already interviewed.

Stakeholder / role	Interest and expected need	Influence / engagement
Oncology clinician (simulated)	Review monitoring context, alerts, laboratory records and entered decisions.	High / validation walkthrough proposed
Nurse / treatment-unit staff (simulated)	Enter symptoms; inspect relevant observations and alert state.	High / workflow feedback proposed
Patient (synthetic persona)	Accurate association of observations with a patient/session; confidentiality.	High interest / no patient recruited
Data Engineering team (6 members)	Build reliable pipelines, model scores, displays and reproducible tests.	High / internal workshop
DEPI supervisor/assessor	Review scope, technical expectations and deliverable format.	High / review/sign-off not yet documented

2. Plan the Approach

Objectives: (a) integrate time-stamped observation sources; (b) validate, deduplicate and store events; (c) detect unusual five-minute heart-rate/temperature patterns using Isolation Forest; (d) maintain independent rule-based alerts; and (e) show trends, laboratory context and review history. Training and calibration occur offline; a Python worker applies the saved model to new feature windows.

Elicitation plan: hold a six-member requirements workshop using the existing project brief; perform a supervisor walkthrough of a low-fidelity dashboard and event-flow sketch; if an oncology nurse/clinician is available, ask targeted questions about review workflows and information display. Interviews or approvals must be recorded only after they occur. No completed stakeholder interview is claimed in this draft.

Elicitation item	Planned output / owner
Project brief and prior planning drafts	Initial scope / Member 1; stages and risks / Member 2; all members contribute.
Academic supervisor walkthrough	Corrections and approval decision / all six members; rotating coordinator.
Simulated clinician/nurse role review	Interface questions / Member 5; alert/review-record questions / Member 6.
Technical feasibility session	Sources/transport M1; streaming M2; database/features M3; AI M4; interface M5; rules/reviews M6.
Joint review and traceability	Requirements compiled by Members 5-6; each owner supplies tests and evidence.

3. Conduct Elicitation

Current status: the team-derived elicitation baseline is documented here using the project brief, the supplied official guidelines and scope decisions. No patient/clinical-staff interviews or academic supervisor sign-off have been supplied, so the following questions remain proposed for future verification:

- Which observations and lab results should be visible before, during and after a treatment session, and who enters each one?
- Who reviews a new alert, what information explains it, and how is a reviewed alert closed?
- What clinical demonstration rules and data-freshness conditions may be shown safely with synthetic data?
- What fields are required when a reviewer records a treatment-plan or dose-change decision?
- Which software and submission constraints are mandatory according to the DEPI supervisor?

4. Document Observations and Findings

The following requirements represent a proposed solution baseline. “Shall” describes intended behaviour, not completed functionality. The definitions distinguish functional requirements, non-functional targets and project constraints.

4.1 Functional Requirements

ID	Requirement / expected behaviour	Priority
FR-01	Generate repeatable synthetic heart-rate/temperature events, nurse-entered pain/nausea, and ANC/selected CBC laboratory reports, linked to patient and session IDs.	Must
FR-02	Accept Kafka events through Spark, validate schema, units, identifiers and source timestamps; persist accepted data and rejected raw messages with reason codes.	Must
FR-03	Handle duplicate IDs idempotently and preserve older/late events without replacing newer observations on the display.	Must
FR-04	Display patient and session views with trends, latest observation time, lab specimen/result times, missing-data and freshness indicators.	Must
FR-05	Build five-minute per-patient/session heart-rate and temperature feature windows (mean, standard deviation and last-minus-first change for each metric).	Must
FR-06	Train on normal-pattern synthetic training data; calibrate using separate patients; save the Isolation Forest, cutoff, feature version and evidence. Keep test labels out of model inputs.	Must
FR-07	Display AI anomaly flags with scores as unusual-pattern indicators, not clinical diagnoses; show an explicit not-scored reason when data are insufficient.	Must
FR-08	Evaluate independent, versioned rule-based demonstration alerts; rules remain available when AI is down.	Must
FR-09	Record alert provenance, evidence IDs, rule/model version, time, review status and associated clinician-entered decisions.	Must
FR-10	Append simulated reviewer treatment-plan or dose-change entries; retain reason, reviewer ID, time and previous versions. No automated dose calculation.	Must
FR-11	Track received/accepted/rejected/duplicate counts, processing failures and restart/replay behaviour.	Must
FR-12	Defer full chemotherapy-session appointment scheduling beyond the core baseline; consider it only after core delivery.	Could

4.2 Data and Constraints

Data group	Minimum details / handling
Identity	event_id, patient_id, session_id, source_id, event_type, schema_version; original event ID must persist on retry.
Vital sign	metric = heart_rate or temperature; numeric value, agreed unit, measured_at, published_at.
Nursing report	pain or nausea value and defined demonstration score, recorded_at and simulated staff/source identifier.
Laboratory	ANC and selected CBC metrics, explicit unit, specimen_collected_at, result_reported_at and published_at; never infer ANC from vital signs.
Pipeline	received_at, stored_at; reject invalid/malformed records and retain reason; store times as timezone-aware UTC.
AI	feature_window_start/end, six features, scoring coverage/eligibility, model_version, anomaly_score, cutoff, scored_at and evidence record IDs.
Alerts and reviews	alert_id, origin (rule/AI), trigger evidence, version, created_at, status; append-only review decisions with actor/time/reason.

Constraints: use synthetic patient identifiers and a local demonstration environment; no patient-facing home app, real hospital integration, automatic therapy recommendations or complete scheduling module. Results from model validation are engineering results on synthetic scenarios, not evidence of medical effectiveness.

4.3 Non-Functional Requirements

ID	Proposed measurable quality target
NFR-01	In a separate load test: 10 synthetic patients, 2 valid events/second total, 10 minutes. Reconcile all 1,200 acknowledged events and report any missing/lost records.
NFR-02	Target publication-to-database P95 at most 5 seconds in the declared local setup; keep original observation age separate from pipeline latency.
NFR-03	Target appearance of five preselected valid events in the active dashboard within 10 seconds each during normal operation.
NFR-04	After stopping the Spark consumer for 30 seconds while publishing, target reconciliation of the backlog within 60 seconds of restart in a controlled test.
NFR-05	Use 30 synthetic patients split 18/6/6 for training/calibration/test. Save patient splits, generator seeds and scenario labels separately; freeze test patients before tuning.
NFR-06	Report precision, recall, F1, false anomaly episodes per patient-hour, scoring coverage and onset-to-detection delay on held-out synthetic scenarios.
NFR-07	Use an identical feature formula offline and online, preserve model/threshold version, and show failure/insufficient-data status without blocking independent rule alerts.
NFR-08	Provide a documented local setup and synthetic fixtures another team member can reproduce; avoid storing real patient identifiers or secrets in code/logs.

5. Analyze and Prioritize

MoSCoW is used because the supplied guide recommends explicit prioritization. The minimum demonstration requires event ingestion, Spark validation, storage, Isolation Forest scoring, separate rule alerts, dashboard views, and review history. More complex scheduling and external device/mobile integrations are deliberately excluded from the baseline.

Priority	Included capability	Reason
Must have	FR-01 to FR-11, plus minimum data-quality and AI tests	Essential to demonstrate the agreed data-engineering + AI scope.
Should have	Exportable audit summary, configurable dashboard filters and pre-recorded recovery demonstration	Useful for review and assessment but not required for the first vertical slice.
Could have	Deferred scheduling (FR-12), extra vital-sign metrics and additional visual summaries	Not committed; consider only after the core deliverables are complete.
Won't have (this project)	Home monitoring app, real EHR/device connection, clinical-grade risk/dosing recommendation	Outside the selected scope and available evidence.

Key dependencies: event IDs and session mapping before independent modules; accepted-data persistence before AI scoring; patient split before model fit; feature calculations before training; threshold calibration before frozen-test evaluation.

6. Verify and Validate

Plan a walkthrough with the supervisor and team to confirm each requirement's meaning and priority. Record a dated approval decision or required corrections; no supervisor sign-off has yet been supplied. The following acceptance scenarios are planned, not completed:

Check	Scenario / expected evidence	Requirements
AC-01	Publish valid events for all three observation sources (simulated devices, nursing entries and laboratory reports); reconcile accepted records by event ID and patient/session.	FR-01/02/04
AC-02	Test missing fields, invalid units, unknown patient/session and timestamps; ensure rejection reasons and no invalid alerts.	FR-02
AC-03	Replay 50 event IDs; confirm no duplicate observations or alerts; test one conflicting reused ID.	FR-03/09
AC-04	Deliver older after newer data and a missing lab result; confirm history, correct latest reading and explicit unavailable status.	FR-03/04
AC-05	Hand-check five-minute features, window isolation and insufficient-coverage handling.	FR-05/07; NFR-07
AC-06	Fit on training patients, calibrate the cutoff separately, then evaluate untouched test patients; report precision, recall, F1, false flags, coverage and detection delay.	FR-06/07; NFR-05/06
AC-07	Remove the model artifact; confirm AI is unavailable visibly while independent rule alerts and ingestion continue.	FR-07/08; NFR-07
AC-08	Create one alert and two recorded reviewer decisions; inspect full evidence/history after application restart.	FR-09/10
AC-09	Run 1,200 events; reconcile records and measure P95 latency and five-event dashboard visibility. Separately test the 30-second interruption and 60-second recovery target.	FR-02/11; NFR-01–04
AC-10	A second member reproduces setup and tests from the runbook; verify synthetic identifiers, versioned artifacts and absence of credentials in code/logs.	NFR-08

7. Manage Changes

Baseline control: retain versioned requirements, schemas, model/features, supervisor comments and a requirement-to-test register. Changes record the rationale, individual owner, affected deadlines/tests, peer reviewer and approval status. This backup changes ownership only; technical requirements and acceptance criteria remain unchanged.

Requirement group	Owner / peer reviewer	Primary evidence
Scope and first-stage documents	M1-2: planning; M3-4: literature; M5-6: requirements	All six cross-review; supervisor approval remains pending.
Simulation and Kafka	M1 / M2	FR-01; transport part of FR-02/11; labels/splits for NFR-05; AC-01/03/09.
Spark and data quality	M2 / M3	FR-02/03/11 with M3 storage helpers; AC-01 to AC-04 and AC-09.
Database and features	M3 / M4	Storage integrity for FR-02/03/09/10; FR-05; AC-01/03/04/05/08.
Model and scoring	M4 / M5	FR-06; AI output for FR-07/09; NFR-05/06/07; AC-05/06/07.
Dashboard and forms	M5 / M6	FR-04/07; interface part of FR-09/10; AC-04/07/08; display test in AC-09.
Rules and review records	M6 / M1	FR-08/09/10; AC-03/07/08; uses M3 storage and serves M5 forms.
Shared integration	All members / rotating lead	Each owner tests/documents the module; jointly complete AC-01 to AC-10.